Doubly stochastic scaling for the detection of block structures in sparse matrices

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FAU - Erlangen

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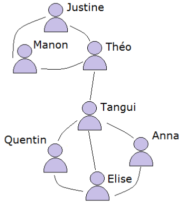
[3]IRIT, Université de Toulouse

Matrix is a powerful tool for **representing data**.

(Social) networks:



⇓

	Anna	*Elise*	*Justine*	*Manon*	*Quentin*	*Tangui*	*Théo*
Anna	♪	X				X	
Elise	X	♪			X	X	
Justine			♪	X			X
Manon			X	♪			X
Quentin		X			♪	X	
Tangui	X	X			X	♪	X
Théo			X	X		X	♪

Interactions between data (2D points):

⇓

Data (genes) according to some parameters:

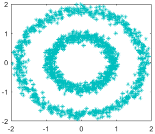
Matrix is a powerful tool for **representing data**.

(Social) networks:

⇓

	Anna	Elise	Justine	Manon	Quentin	Tangui	Théo
Anna	♪	X				X	
Elise	X	♪			X	X	
Justine			♪	X			X
Manon			X	♪			X
Quentin		X			♪	X	
Tangui	X	X			X	♪	X
Théo			X	X		X	♪

Interactions between data (2D points):



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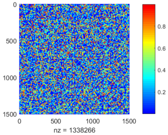


Data (genes) according to some parameters:

Matrix is a powerful tool for **representing data**.

(Social) networks:

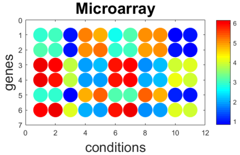
⇓

	Anna	Elise	Justine	Manon	Quentin	Tangui	Théo
Anna	♪	X				X	
Elise	X	♪			X	X	
Justine			♪	X			X
Manon			X	♪			X
Quentin		X			♪	X	
Tangui	X	X			X	♪	X
Théo			X	X		X	♪

Interactions between data (2D points):

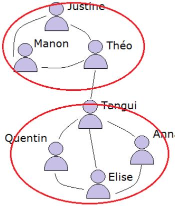
⇓

Data (genes) according to some parameters:



Block structures may convey **important information** about the matrix.

(Social) networks:



⇓

	Manon	Justine	Théo	Tangui	Quentin	Elise	Anna
Manon	♪	X	X				
Justine	X	♪	X				
Théo	X	X	♪	X			
Tangui			X	♪	X	X	X
Quentin				X	♪	X	
Elise				X	X	♪	X
Anna				X		X	♪

Interactions between data (2D points):

⇓

Data (genes) according to some parameters:

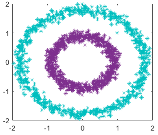
⇓

Block structures may convey **important information** about the matrix.

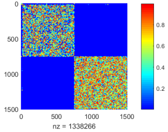
(Social) networks:

$\Downarrow$

Interactions between data (2D points):



$\Downarrow$



Data (genes) according to some parameters:

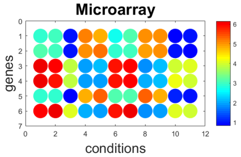
$\Downarrow$

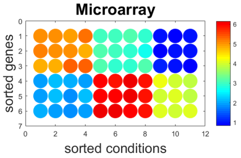
Block structures may convey **important information** about the matrix.

(Social) networks:

Interactions between data (2D points):

Data (genes) according to some parameters:





How to detect block structures in matrices?

- The **bi-stochastic scaling** highlights the block structure.
- The **spectral analysis** of bi-stochastic scaling enables to detect this structure.

Definition: Given $A \in \mathbb{R}_+^{n\times n}$, finding two positive vectors $r, c \in {\mathbb{R}_+^*}^n$ such that:

$$P \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} r_1 & & \\ & \ddots & \\ & & r_n \end{bmatrix} A \begin{bmatrix} c_1 & & \\ & \ddots & \\ & & c_n \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

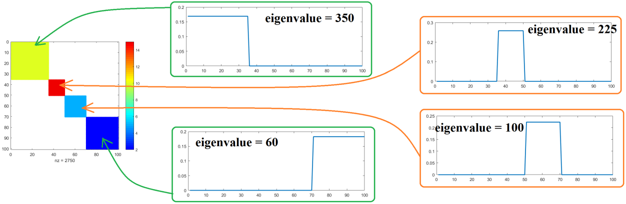
$$P^T \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} c_1 & & \\ & \ddots & \\ & & c_n \end{bmatrix} A^T \begin{bmatrix} r_1 & & \\ & \ddots & \\ & & r_n \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

$\iff$ all the row sums and column sums of $P = RAC$ are equal to 1.

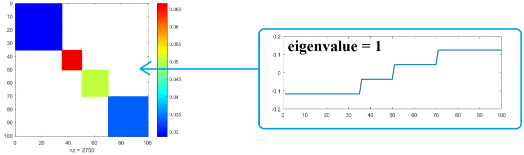
Vectors r and c are called scaling factors of A.

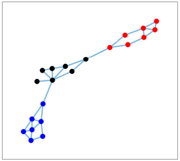
Introduction - Spectral Property of bi-Stochastic Scaling

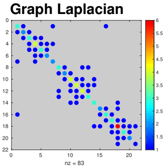
Without scaling, **each block** is characterized by **its own eigenvalue**.
$\Rightarrow$ **Requirement of** 4 **eigenvectors** to detect the 4 blocks.

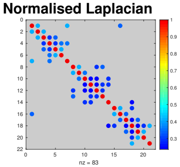


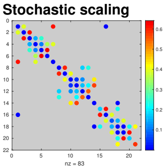
After scaling, **every block** is characterized by the leading **eigenvalue** 1.
$\Rightarrow$ **One unique eigenvector** is enough to detect the 4 blocks.

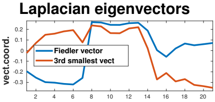


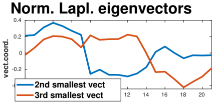


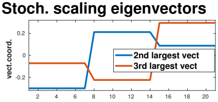

Graph Laplacian
nz = 83

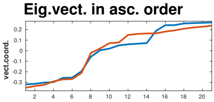

Normalised Laplacian
nz = 83

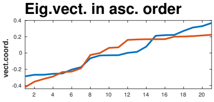

Stochastic scaling
nz = 83

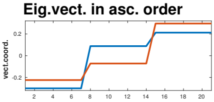

Laplacian eigenvectors
vect.coord.
Fiedler vector
3rd smallest vect

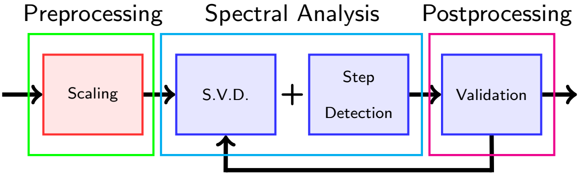

Norm. Lapl. eigenvectors
vect.coord.
2nd smallest vect
3rd smallest vect


Stoch. scaling eigenvectors
vect.coord.
2nd largest vect
3rd largest vect

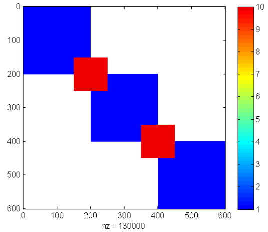

Eig.vect. in asc. order
vect.coord.


Eig.vect. in asc. order
vect.coord.

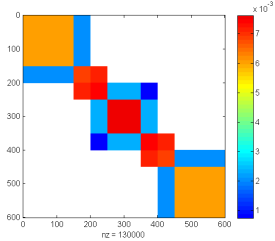

Eig.vect. in asc. order
vect.coord.



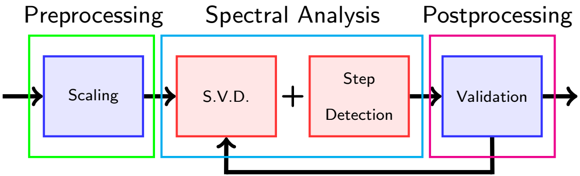
Doubly-stochastic scaling of the matrix to highlight its numerical block structure.



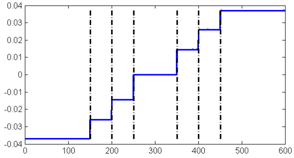
$\Longrightarrow$



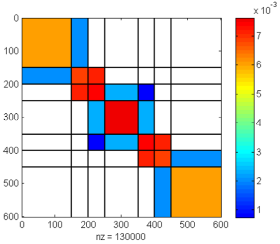
Our Algorithm - Main Steps



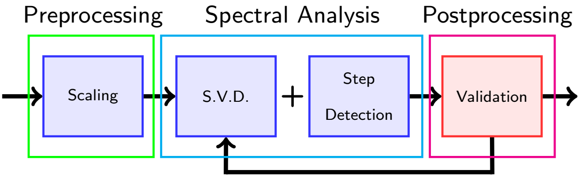
Dominant singular vectors (potentially just 1 couple would be enough) allow to detect these blocks.



$\Longrightarrow$

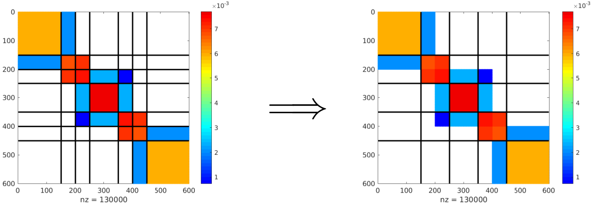


Our Algorithm - Main Steps



If several iterations :

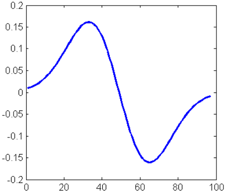
- Clustering overlapping.
- Amalgamation of some blocks.
- Convergence test.



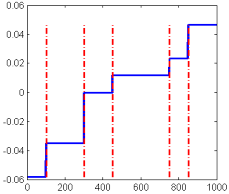
Our Algorithm - Block Detection as a Signal Processing Problem

- Canny's filters can detect steps in signals[1].
- Singular vector = Signal out of which we aim to detect the steps.
- Convolution product between a signal and a filter.

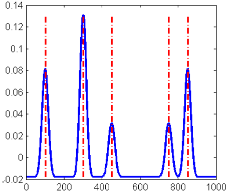
⇒ Maxima correspond to steps in the signal.



filter



vector



convolution

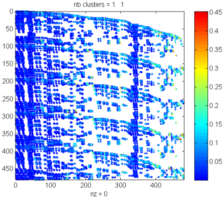
[1] J. Canny, *A Computational Approach to Edge Detection*, 1986

Our Algorithm - Number of Blocks

- Block detection process is applied on each vector at our disposal.
- We can iterate and incorporate new set of vectors.

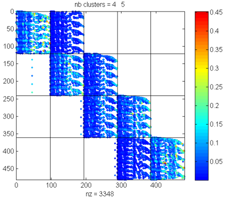
$\Rightarrow$ The number of blocks can blow up quickly.

matrix `rbsb480`[2]



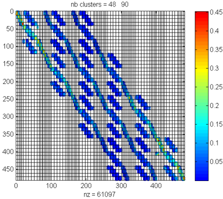
initial matrix.

$\Rightarrow$



after the 1st ite.

$\Rightarrow$



after the 3rd ite.

$\Rightarrow$ Process of block amalgamation is applied after each iteration.

[2] T. Davis, Y. Hu, *The University of Florida sparse matrix collection*, 2011

Our Algorithm - Modularity

Modularity measure based on Newman and Girvan's[3]:

$$\mathcal{Q}_r = \frac{1}{n} \sum_{k=1}^{r_c} \left(v_k^T A A^T v_k - \frac{1}{n} |C_k|^2 \right) \geq 0$$

Doubly-stochastic property of A $\Rightarrow$ homogenisation of several modularity measures.

Process of block amalgamation:

- Amalgamation of the pair of blocks that maximises the increase of modularity.
- Loop until local maximum is reached.
- No pairwise amalgamation improvement $\Rightarrow$ No modularity increase is still possible, whatever the number of amalgamated blocks.

[3] M. Newman, *Analysis of weighted networks*, 2004

Non-linear upper bound of the modularity:

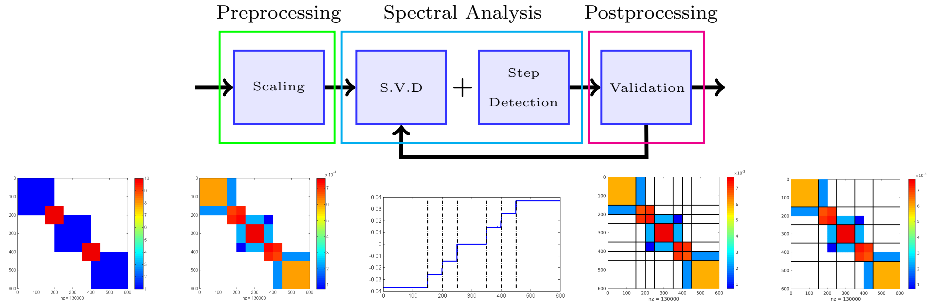
$$\mathcal{Q}_r \leq 1 - \frac{1}{nb_{Clust}} = \mathcal{K}_r.$$

Calling ρ the ratio:

$$\rho = \frac{\mathcal{Q}_r^{upd} - \mathcal{Q}_r^{ref}}{\mathcal{K}_r^{upd} - \mathcal{K}_r^{ref}} = \frac{\mathcal{Q}_r^{upd} - \mathcal{Q}_r^{ref}}{\frac{1}{{nb_{clust}}^{ref}} - \frac{1}{{nb_{clust}}^{upd}}},$$

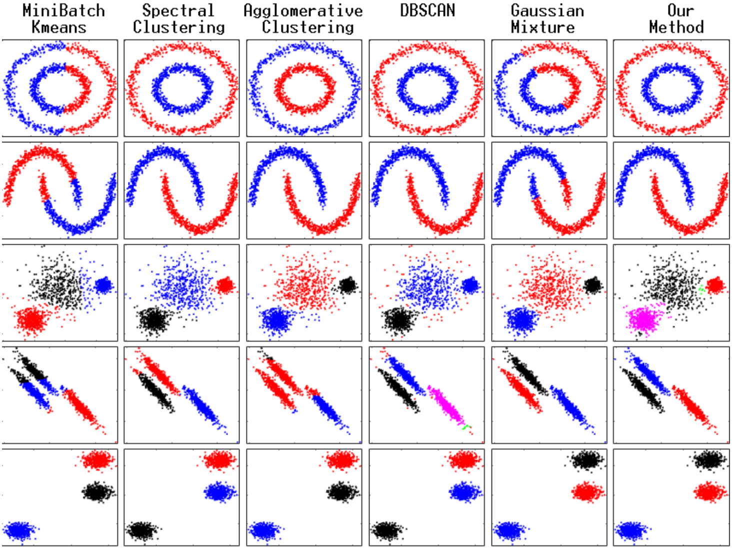
A new clustering is accepted iff $\rho > \epsilon$, with $\epsilon \in [0.01, 0.1]$ a threshold.

$\Rightarrow$ **A new clustering is accepted only if the increase of modularity is within a fraction of the asymptotic difference of the upper bounds.**



- On a perfect block structure, the **doubly-stochastic scaling** highlights the **whole matrix structure** on **only one pair of singular vectors** (a few pairs in non perfect cases).
- With the use of **tools from signal processing**, **no information** about the clustering (number of blocks, etc.) is **required by our method**.
- The **Newman and Girvan modularity** is used to **control the number of blocks** and **derive a stopping criterion**.

Applications - Shape Detection



Applications - Shape Detection

NMI	MiniBatch Kmeans	Spectral Clustering	Agglomerative Clustering	DBSCAN	Gaussian Mixture	Our Method
2Circles	2.9×10^{-4}	**1**	0.993	**1**	1.3×10^{-6}	**1**
2Moons	0.39	**1**	**1**	**1**	0.401	**1**
3Variances	0.813	0.915	0.898	0.664	**0.942**	0.902
3Aniso.	0.608	0.942	0.534	0.955	**1**	0.996
3Points	**1**	**1**	**1**	**1**	**1**	**1**

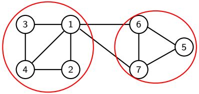
NMI (Normalised Mutual Information) : a measure $\in [0,1]$ to compare 2 clusterings $\mathcal{C}$ and $\mathcal{K}$ s.t $NMI(\mathcal{C},\mathcal{K}) = 1$ iff $\mathcal{C} = \mathcal{K}$.

Our algorithm:

- is the only one, with *Spectral Clustering*, to return a consistent clustering for every data set.
- is the only one, with *DBSCAN*, that does not require the number of clusters.

Applications - Community Detection (inspired from[5])

- Networks of 500 nodes generated with the LFR Benchmark[4].
- Average degree $\overline{k} \in \{10, 20, 40, 75\}$.
- Mixing parameter μ : $\mu(1) = \frac{k_1^{out}}{k_1^{tot}} = \frac{2}{5}$
  big $\mu \Rightarrow$ weak community structure.

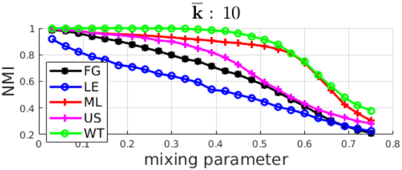


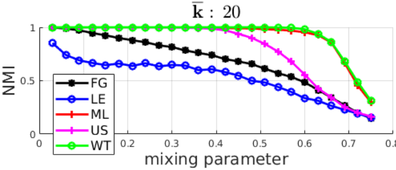
Comparison of 4 algos from the *igraph* package :	Based on :
Leading Eigenvector [LE]	Newman's modularity, Spectral tools.
Louvain's [ML]	Newman's modularity optimisation.
Fast&Greedy [FG]	Newman's modularity optimisation.
Walktrap [WT]	Hierarchical clustering, Newman's modularity.

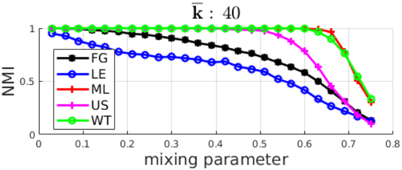
[4] A. Lancichinetti et al.,*Benchmark graphs for testing community detection algorithms*, 2008

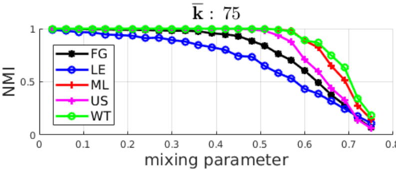
[5] Z. Yang et al.,*A Comparative analysis of community detection algorithms on artificial networks*, 2016

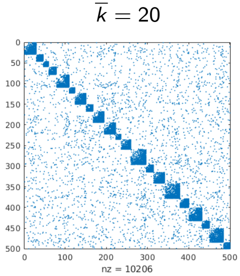
Applications - Community Detection

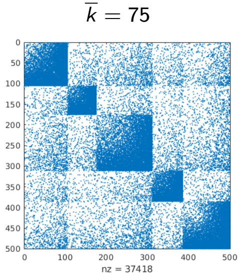




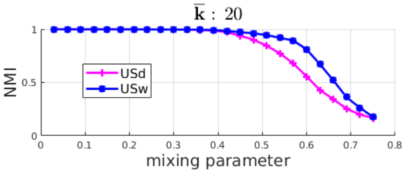


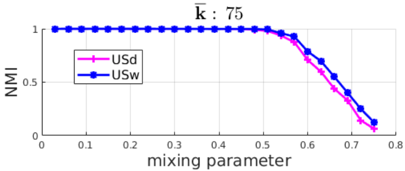






USd works on A, USw works on $A + \eta e e^T, \eta = 0.15$

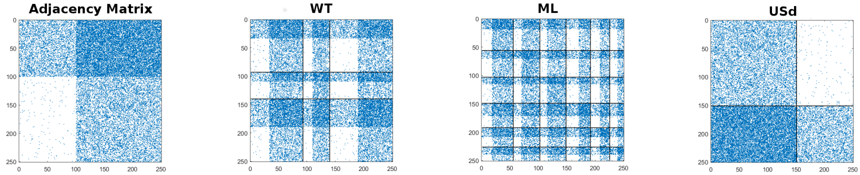




Our algorithm is

- Totally unsupervised.
- Versatile : it is competitive (for community detection), if not superior (for shape detection) in comparison with on-purpose algorithms.

+ Able to detect communities in a graph with flows:



One of the best community detection algorithm[6].

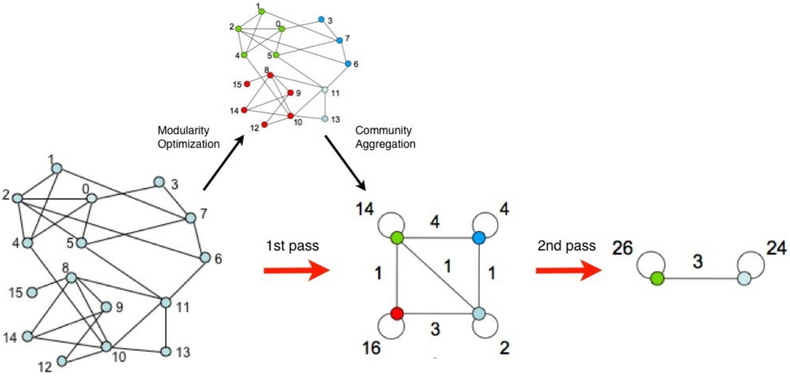


Figure: Example of Louvain steps from [7]

[6] Yang et.al., *A comparative analysis of com. detect. alg. on artificial networks*, 2016
[7] Blondel et.al., *Fast unfolding of communities in large networks*, 2008

Louvain algorithm:

- ☺ is accurate,
- ☺ has a good computing time,
- ☺ does not require to set up parameters,
- ☺ has been generalised for other modularity measures[8],
- ☹ has difficulties with unbalanced size communities,
- ☹ has difficulties with directive graphs with flows.

[8] Campigotto et.al., *A generalized and adaptative method for com. detect.*, 2014

Modularities & bi-Stochasticity - Louvain

Louvain algorithm:

- ☺ is accurate,
- ☺ has a good computing time,
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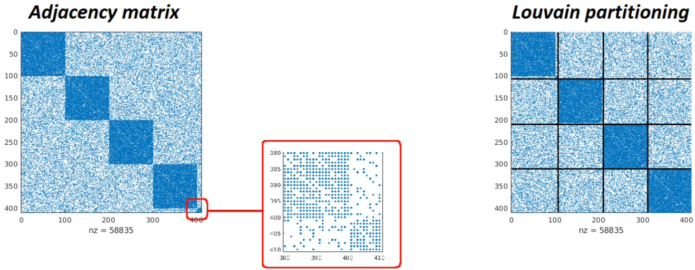

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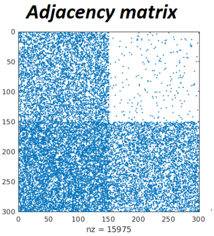

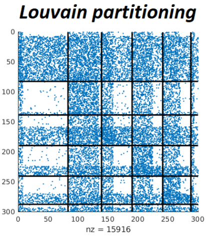

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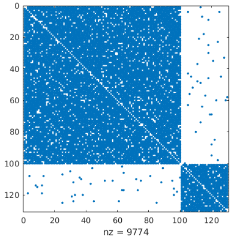
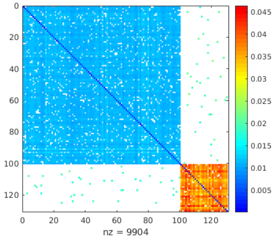
$\Longrightarrow$ We propose to use **the doubly-stochastic scaling** as a **preprocessing for Louvain**, to **improve the detectability of communities**.

[8] Campigotto et.al., *A generalized and adaptative method for com. detect.*, 2014

Communities $\mathcal{C}_1$ & $\mathcal{C}_2$ s.t $|\mathcal{C}_1| = n_1 > |\mathcal{C}_2| = n_2$; $p_{in} = 0.9$; $p_{out} = 0.01$.





Average amount of edges between a node and nodes from its community:

Before scaling:

$$☹ \begin{cases} \frac{1}{n_1} \sum_{i\in\mathcal{C}_1} \sum_{j\in\mathcal{C}_1} A(i,j) = 89 \\ \frac{1}{n_2} \sum_{i\in\mathcal{C}_2} \sum_{j\in\mathcal{C}_2} A(i,j) = 26.5 \end{cases}$$

After scaling:

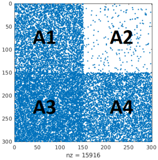
$$☺ \begin{cases} \frac{1}{n_1} \sum_{i\in\mathcal{C}_1} \sum_{j\in\mathcal{C}_1} P(i,j) = 0.99 \\ \frac{1}{n_2} \sum_{i\in\mathcal{C}_2} \sum_{j\in\mathcal{C}_2} P(i,j) = 0.98 \end{cases}$$

Theoretically:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \implies P \to \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \text{ with } \begin{cases} r \to (0, +\infty)^T \\ c \to (+\infty, 0)^T \end{cases}$$

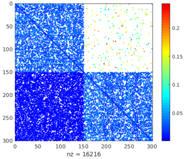
Practically:

if $A =$



with $A_1, ..., A_4 \in \mathbb{R}^{n \times n}$ and $\begin{cases} \frac{1}{n} \sum_{i,j} A_1(i,j) = 30 \\ \frac{1}{n} \sum_{i,j} A_4(i,j) = 30 \\ \frac{1}{n} \sum_{i,j} A_2(i,j) = 1.5 \\ \frac{1}{n} \sum_{i,j} A_3(i,j) = 45 \end{cases}$

then $P =$

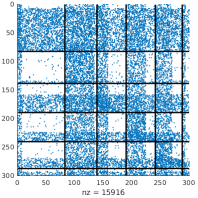


with $\begin{cases} \frac{1}{n} \sum_{i,j} P_1(i,j) = \frac{1}{n} \sum_{i,j} P_4(i,j) = 0.8 \\ \frac{1}{n} \sum_{i,j} P_2(i,j) = \frac{1}{n} \sum_{i,j} P_3(i,j) = 0.2 \end{cases}$

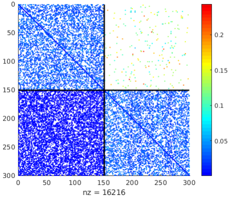
Modularities & bi-Stochasticity - Louvain on scaled matrices

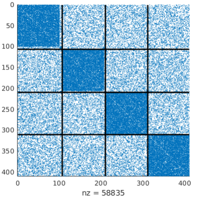
☹ Louvain's results when applied on raw matrices: ☹

☺ Louvain's results when applied on scaled matrices: ☺



$\Longrightarrow$





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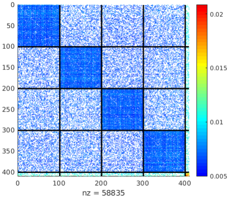


- Louvain's algorithm can be generalised to other modularity measures[9].
- A list of modularity measures are provided by the authors.
- These measures have generally been designed for specific graphs.

$\Rightarrow$ We propose to investigate a generalisation to doubly-stochastic graphs for some of these measures.

We investigate 4 measures applied on a given adjacency matrix A:

- Newman&Girvan modularity $F_{NG}(X)$
- Zahn modularity $F_Z(X)$
- Balanced modularity $F_{BM}(X)$
- Correlation clustering $F_{CC}(X)$

[9] Campigotto et.al., *A generalized and adaptative method for com. detect.*, 2014

- $\mathcal{G} = (V, E)$ is an undirected network of n vertices.
- $A \in \mathbb{R}_+^{n \times n}$ is s.t. $a_{i,j} = \begin{cases} \text{weight of edge } (i,j) \text{ if } (i,j) \in E, \\ 0 \text{ otherwise.} \end{cases}$
- $d_k = \sum_{i=1}^{n} a_{k,i}$ is the degree of node k and $m = \frac{1}{2} \sum_k d_k$.
- If $\mathcal{G}$ is unweighted, $a_{i,j} \in \{0, 1\}$, and $\overline{A}$ is s.t. $\overline{a}_{i,j} = 1 - a_{i,j}$.
- $X \in \{0, 1\}^{n \times n}$ s.t. is $x_{i,j} = 1 \iff i$ and j belong to the same community.
- $\overline{X}$ is s.t. $\overline{x}_{i,j} = 1 - x_{i,j}$.

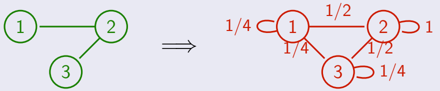
Modularities & bi-Stochasticity - Newman & Girvan modularity[10]

Principle: Random graphs do not have community structure.

$$F_{NG}(A, X) = \sum a_{i,j} x_{i,j} - \sum b_{i,j} x_{i,j}$$

with $b_{i,j} = d_i d_j / 2m$, and $2m = \sum_{i \in V} d_i$:

$d_1 = 1;\ d_2 = 2;\ d_3 = 1$ $\qquad$ $d_1 = 1;\ d_2 = 2;\ d_3 = 1$



Doubly-stochastic graphs.

A is **doubly-stochastic** $\implies$ $2m = n$ and $\forall i, d_i = 1$, and we have:

$$F_{NG}(A, X) = \sum_{i,j \in V} (a_{i,j} - 1/n) x_{i,j}$$

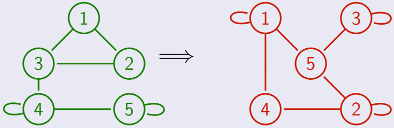
[10] Newman, *Analysis of weighted networks*, 2004

Principle: Random graphs do not have community structure.

$$F_{BM}(A, X) = F_{NG}(A, X) + F_{NG}(\overline{A}, \overline{X})$$

with $F_{NG}(A, X) = \sum_{i,j \in V} \left(a_{i,j} - d_i d_j / 2m\right) x_{i,j}$

$$\overline{a_{i,j}} = 1 - a_{i,j} \implies \begin{cases} \overline{d_i} = n - d_i \\ \overline{2m} = n - 2m \end{cases}$$



Doubly-stochastic graphs.

If A **doubly-stochastic**: $\overline{a_{i,j}} = 1 - a_{i,j} \implies \overline{d_i} = n - 1, \overline{2m} = n(n-1)$

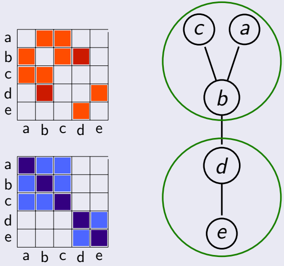
$$\implies F_{BM}(A, X) = 2F_{NG}(A, X)$$

[11] Conde-Céspedes, Marcotorchino, *Comparison of linear modularization criteria of networks using relational metric*, 2013

Principle for unweighted graphs.

$\mathcal{A} \subset V \times V$:

$\mathcal{X} \subset V \times V$



$$\begin{cases} \mathcal{A} = (\mathcal{A} \cap \mathcal{X}) \cup (\mathcal{A} \cap \overline{\mathcal{X}}) \\ \mathcal{X} = (\mathcal{X} \cap \mathcal{A}) \cup (\mathcal{X} \cap \overline{\mathcal{A}}) \end{cases}$$

Zahn's distance: $d_Z(\mathcal{A}, \mathcal{X}) = |\overline{\mathcal{X}} \cap \mathcal{A}| + |\overline{\mathcal{A}} \cap \mathcal{X}|$

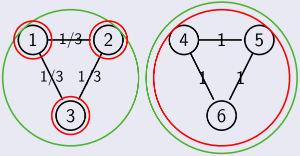
$\iff$ Zahn's modularity :

$$F_Z(A, X) = \sum (a_{i,j} x_{i,j} + \bar{a}_{i,j} \bar{x}_{i,j})$$

[12] Zahn, *Approximating symmetric relations by equivalence relations*, 1964

Generalisation to weighted graphs.

- Proposition from[13]: $\bar{a}_{i,j} = a_{max} - a_{i,j}$.
- Our proposal: $\bar{a}_{i,j} = a_{mean} - a_{i,j}$.



From [13]: $F_Z(A, X) < F_Z(A, X)$. ☹

Us: $F_Z(A, X) > F_Z(A, X)$. ☺

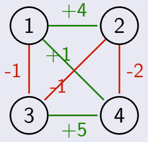
Doubly-stochastic graphs.

If A is **doubly-stochastic**, then $a_{mean} = \frac{1}{n}$, and we can consider

$$F_Z(A, X) = \sum_{i,j \in V} (a_{i,j} - 1/2n) x_{i,j}$$

[13] Campigotto et.al.'A generalized and adaptative method for com. detect.' (2014)

Principle for signed graphs.

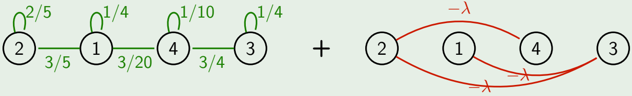


$$\text{Similarities}: \; E^+ = \{(1,2),(1,4),(3,4))\} = \{(i,j) : a_{i,j} > 0\}$$

$$\text{Dissimilarities}: \; E^- = \{(1,3),(2,3),(2,4)\} = \{(i,j) : a_{i,j} < 0\}$$

$$\Longrightarrow F_{CC}(A,X) = \sum_{(i,j)\in E^+} a_{i,j}x_{i,j} - \sum_{(i,j)\in E^-} a_{i,j}\overline{x}_{i,j}$$

Doubly-stochastic case.



$$\Longrightarrow F_{CC,\lambda}(A,X) = \sum_{(i,j)\in E} a_{i,j}x_{i,j} - \lambda \sum_{(i,j)\notin E} x_{i,j}$$

[14] Demaine, Immorlica, *Correlation clustering with partial information*, 2003

Modularities & bi-Stochasticity - Summary and Comparisons

Generalisation to **bi-stochastic graphs** creates **2 types of parametrized modularities**:

- **Newman&Girvan type:** $F(A, X) = \sum_{i,j} (a_{i,j} - \alpha/n)\, x_{i,j}$.

(∗) *Other measures behave alike on bi-stochastic graphs, often with $\alpha = 1$.*

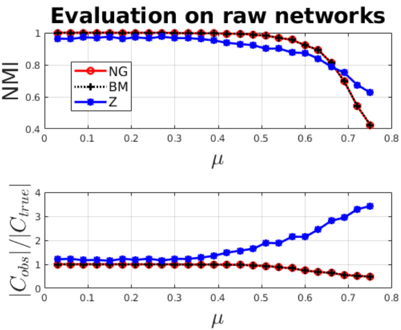
- **Correlation Clustering type:** $F(A, X) = \sum_{(i,j) \in E} a_{i,j} x_{i,j} - \lambda \sum_{(i,j) \notin E} x_{i,j}$.

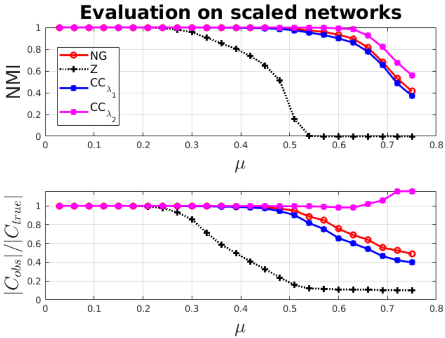
Our tests on LFR benchmark[15]:

- Binary graphs: Newman&Girvan (**NG**), Balanced (**BM**) and Zahn (**Z**) modularities.
- Scaled graphs:
  - Newman&Girvan type with $\alpha = 1$ (**NG**) and $\alpha = 1/2$ (**Z**).
  - Correlation Clustering type with $\lambda = 1/n$ ($\mathbf{CC}_{\lambda_1}$) and $\lambda = 2/n$ ($\mathbf{CC}_{\lambda_2}$)

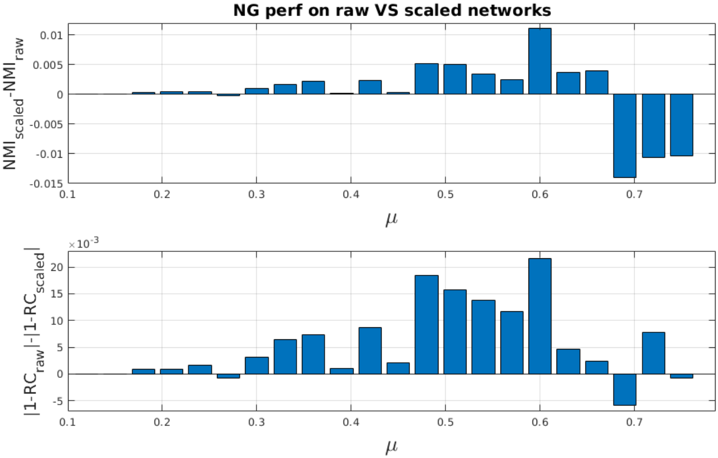
[15] Lancichinetti et.al., *Benchmark graphs for testing com. detect. alg.*, 2008

Modularities & bi-Stochasticity - Comparison of the criteria





- <u>Raw networks</u>: Louvain with NG & BM has very similar accuracy. Z is not competitive.
- <u>Scaled networks</u>: Z is not competitive. NG and CC_{λ_1} have similar behaviours. CC_{λ_2} is better.



- With NG criterion, Louvain applied on scaled networks is more accurate when communities are not "too mixed".

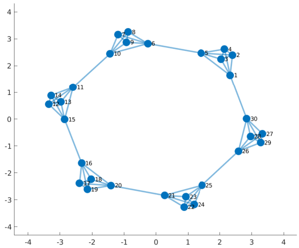
Modularities & bi-Stochasticity - Conclusions

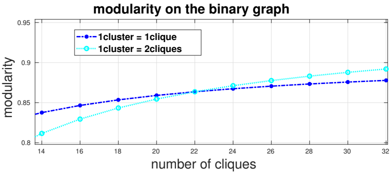
- **Doubly-stochastic scaling unifies some measures** that can be used in Louvain.
- **A range of measures can be derived** from the generalisation of modularities to **doubly-stochastic scaling**. For instance, the NG's type:

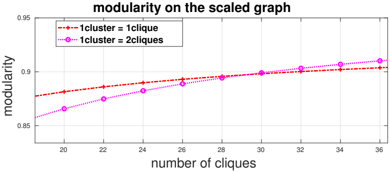
$$\begin{array}{rccc} F: & Eq(n) \times]0,1] & \longrightarrow & \mathbb{R} \\ & (X, K) & \mapsto & \sum\limits_{i,j \in V} (a_{i,j} - K)\, x_{i,j} \end{array}$$

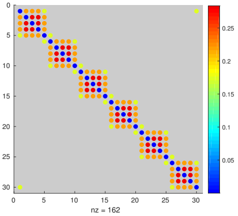
- **Doubly-stochastic scaling** as a preprocessing for Louvain algorithm **improves its accuracy**.
- The **measure used in Louvain** should be chosen carefully, as its accuracy depends on it.

Modularities & bi-Stochasticity - Resolution Limit









Conclusions & Perspectives

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[18] P.Knight, *The Sinkhorn-Knopp algorithm: convergence and applications*, 2008

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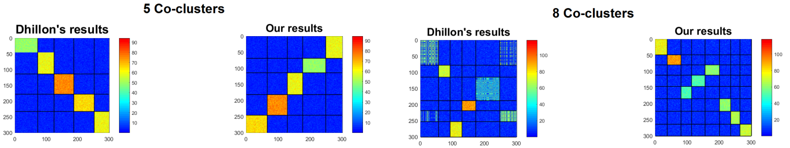

[19] I.Dhillon, *Co-clustering doc. and words using bipartite spectral graph part.*, 2001

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  - TODO Investigating more general families of scaling (**prescribed rows and columns sums**).

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Thank you for your attention